



Selective and Cost-Aware Retrieval Routing for RAG

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Motivation

RAG pipelines apply the same retrieval strategy to every query regardless of difficulty—wasting calls on easy queries and underinvesting on hard ones. We frame per-query retrieval selection as an **offline contextual bandit**: a lightweight policy routes each query to its best retrieval action, making cost explicit through per-call and rewrite penalties.

- Fixed-action baselines (BM25, dense, hybrid) cannot adapt to query-level variation in difficulty or retrieval confidence.
- A learned router should exploit cheap wins and invest extra calls only when retrieval gain justifies cost.
- Unlike LLM-driven routers (Adaptive-RAG, Self-RAG), our policy uses only lightweight BM25 and query features—**no LLM inference at routing time**.
- Primary scope: **retrieval-stage reward**—no LLM fine-tuning. Downstream QA (EM/F1) is shown as an exploratory pilot only ($n = 30$), not a main claim.

Formulation & OPE Framework

Each query is a one-step routing decision:

$$\pi(s) = \arg \max_a \mathbb{E}[R \mid s, a]$$
$$R = \underbrace{\text{Recall@5} + 0.5 \cdot \text{MRR}}_{\text{retrieval quality}} - \underbrace{0.10 \cdot \mathbf{1}[\text{rewrite}]}_{\text{rewrite}} - \underbrace{0.03 \cdot (\text{calls} - 1)}_{\text{call}}$$

- State s** : 14 features—query length, capitalized spans, BM25 top score / gap / entropy, 8 WH-word indicators. No LLM at inference time.
- Actions**: BM25 | dense | hybrid | keyword rewrite; Gemini variants are costly optional extensions.
- Selective policy (main)**: ridge / ExtraTrees on a *full-information* reward table—all four actions' rewards observed per training query. CV-selected: ExtraTrees (SciFact), Ridge (NFCorpus).
- Selected-action bandit (diagnostic)**: only the executed action's reward observed—partial-feedback training. Delta gap (+0.029 SciFact, +0.003 NFCorpus) quantifies the cost of logging bias.

OPE: Doubly Robust estimator (DM + IPS correction)

$$\hat{V}_{\text{DR}} = \underbrace{\frac{1}{n} \sum_i \hat{r}(s_i, \pi(s_i))}_{\text{DM (model-based)}} + \underbrace{\frac{1}{n} \sum_i \frac{\mathbf{1}[a_i = \pi(s_i)]}{p(a_i | s_i)} (r_i - \hat{r}(s_i, a_i))}_{\text{IPS correction (data-based)}}$$

Consistent if *either* $\hat{r}$ or p is correct—doubly robust against model mis-specification.

Logging	Estimator	SciFact err	NFC err
Uniform	DM	0.023	0.052
Uniform	IPS	0.127	0.052
Uniform	DR	0.058	0.038
Sparse	IPS	0.268	0.066
Sparse	DR	0.093	0.068

Abs. error vs. true policy value (target: Selective policy). IPS 2× worse under sparse logging on SciFact (0.127→0.268); DR absorbs coverage gaps—validates full-information reward table.

Implementation Pipeline

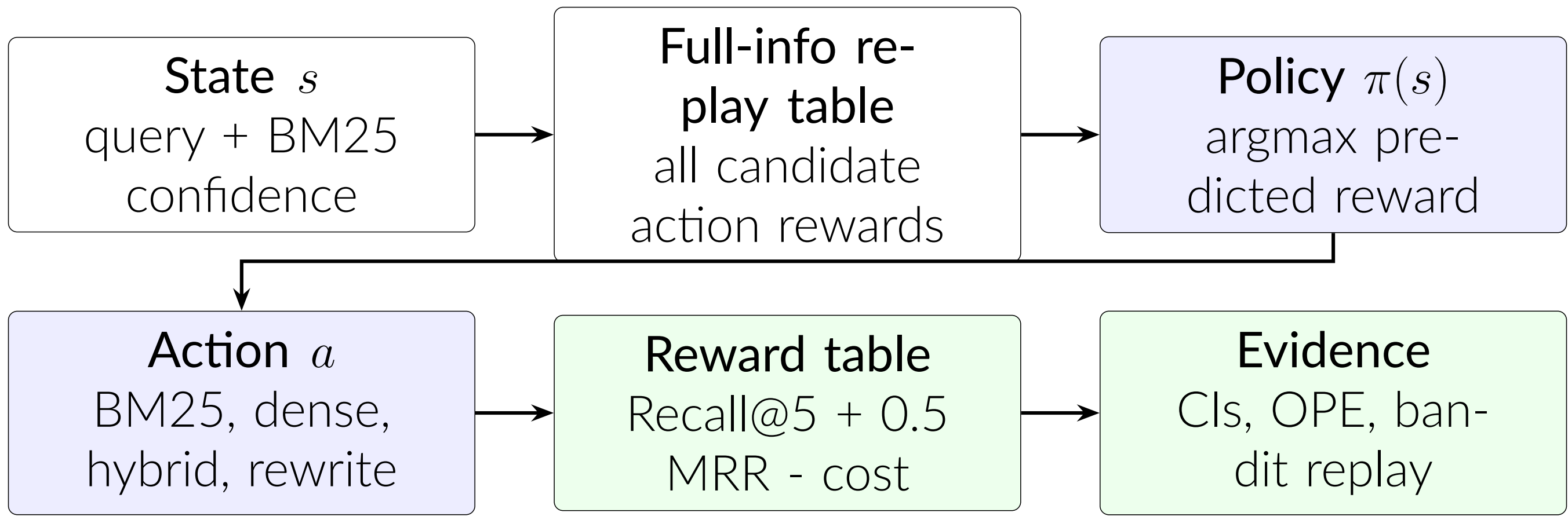
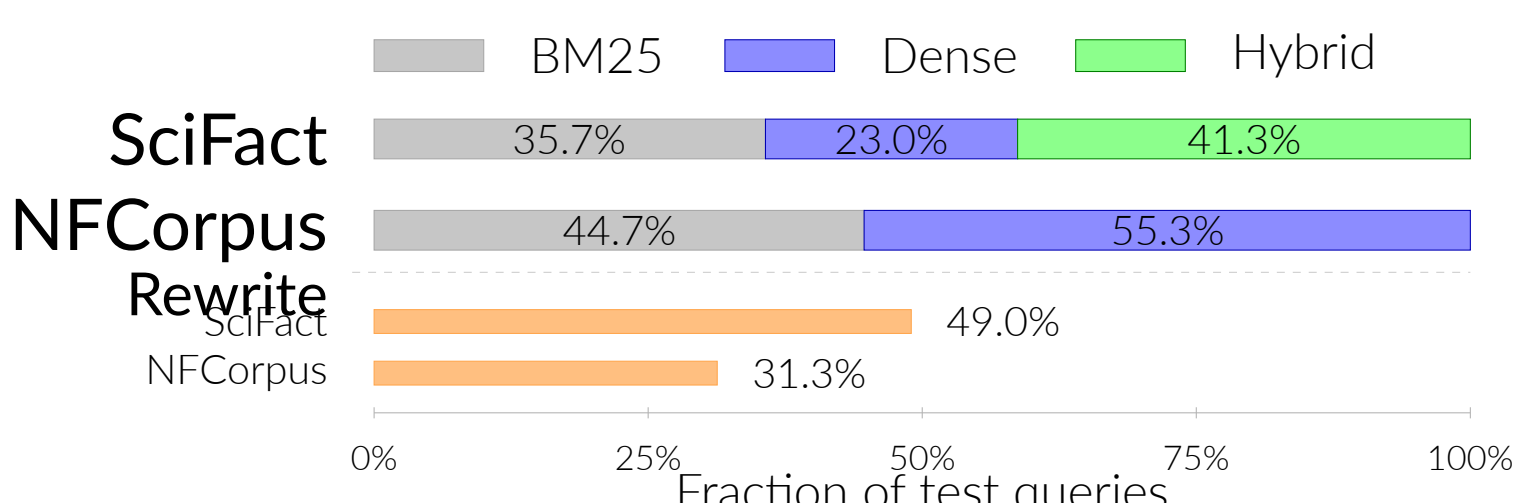


Figure 1. Offline contextual-bandit framing: replayable action rewards train a policy that maps query state to retrieval action.

Policy Action Distribution

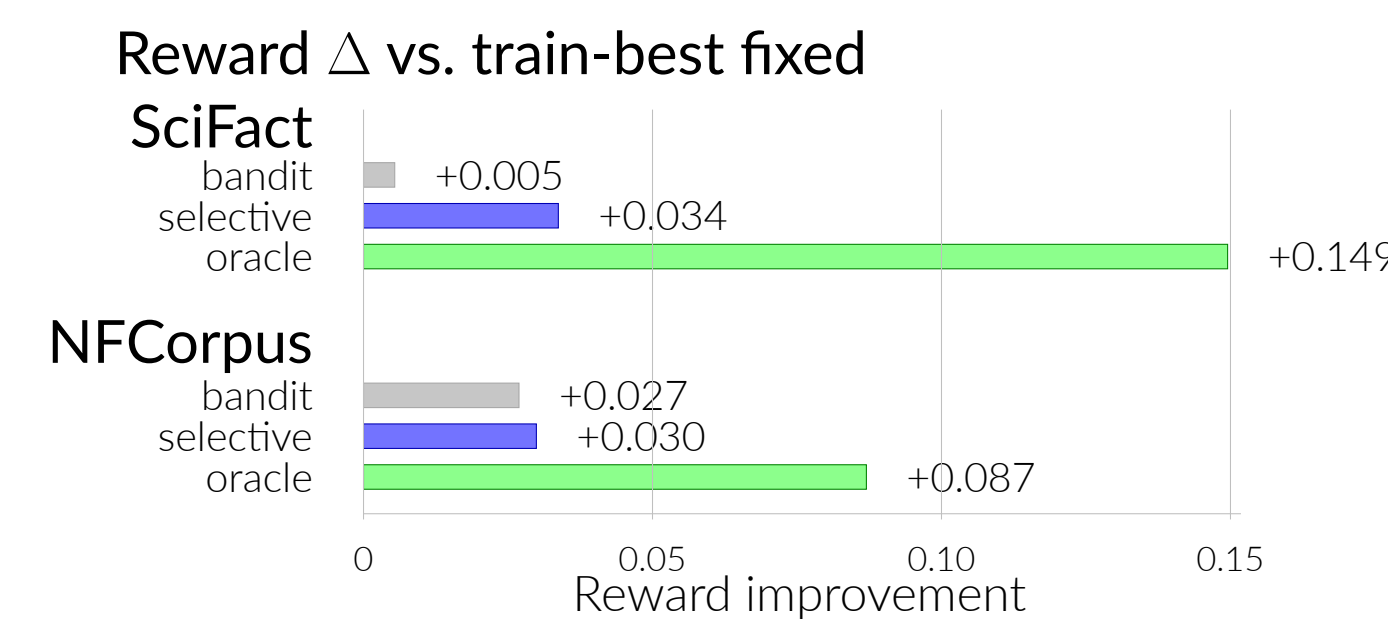


SciFact benefits from hybrid retrieval (41.3%) and frequent rewrites; NFCorpus stays cost-efficient with BM25 or Dense only—the policy **never selects Hybrid on NFCorpus**, consistent with its smaller, domain-specific vocabulary.

Main Results

Dataset	Method	Reward	Δ (95% CI)	Calls/q
SciFact	Train-best fixed	1.057	— (baseline)	2.00
SciFact	Selected-action bandit	1.062	+0.005	1.03
SciFact	Selective policy	1.091	+0.034 [0.009, 0.058]	1.41
SciFact	Oracle upper bound	1.206	+0.150	1.01
NFCorpus	Train-best fixed	0.377	— (baseline)	1.00
NFCorpus	Selected-action bandit	0.404	+0.027	1.14
NFCorpus	Selective policy	0.407	+0.030 [0.005, 0.055]	1.00
NFCorpus	Oracle upper bound	0.464	+0.087	1.02

Reward = Recall@5 + 0.5 · MRR − cost. **Calls/q**: avg. retrieval actions per query (train-best fixed = 2 calls on SciFact). Bootstrap 95% CIs stay above zero on both datasets. Full-corpus BEIR: 600 train / 300 test queries per dataset (SciFact: 5,183 docs; NFCorpus: 3,633 docs).



(a) Reward-delta ladder

Key Results

Metric	SciFact	NFCorpus
Selective Δ	+0.034	+0.030
Oracle gap closed	22.6%	34.4%
95% CI	[0.009, 0.058]	[0.005, 0.055]
Constr. $\Delta^\dagger$	+0.047	+0.033
Calls/query	1.41	1.00
3-seed win rate	0.667	0.667

[†]Constrained Δ is utility vs. a lower utility baseline (1.042 / 0.377); Selective Δ is reward vs. train-best reward. Different objectives & baselines.

(b) Summary

Figure 2. Selective-policy gains with cost-aware and robustness diagnostics.

The selective policy closes **22.6%** of the SciFact oracle gap and **34.4%** of the NFCorpus oracle gap. On NFCorpus, the same reward gain is achieved at **zero extra retrieval cost** (1.00 call/query).

Cross-Dataset Generalization

Routing policies trained on one domain transfer to unseen target domains:

Source	Target	Policy Reward	Target Best	Δ	Calls/q
SciFact	SciFact	1.093	1.057	+0.036	1.48
SciFact	NFCorpus	0.386	0.377	+0.009	1.60
NFCorpus	NFCorpus	0.407	0.377	+0.030	1.00
NFCorpus	SciFact	1.064	1.057	+0.007	1.00

All four transfer settings positive. NFCorpus→SciFact at zero extra cost; SciFact→NFCorpus incurs 1.60 calls/query (hybrid-heavy routing).

Robustness & Extended Analysis

- OPE stability**: bootstrap 95% CIs confirm all main-result deltas separated from zero; DR preferred over IPS (see Formulation).
- Replay regret**: full-info training: regret 67.4 vs. 82.1 (SciFact), 29.6 vs. 43.2 (NFCorpus)—≈20% over partial-feedback.
- Difficulty buckets**: high action-disagreement queries: SciFact +0.049, NFCorpus +0.096—router gains where it matters.
- Budget sweep**: constrained policy ($\lambda = 0.06$) holds +0.039 utility at 1.05 calls/query (SciFact); NFCorpus unchanged.
- 3-seed robustness**: win rate 0.667 both datasets; deltas +0.021/+0.025 (300/150 train/test vs. main 600/300).

Downstream QA Pilot (HotpotQA, $n = 30$, Gemini reader)

Retrieval	EM	F1	Calls/q
BM25	0.733	0.823	1.0
Train-best fixed	0.767	0.852	2.0
Learned policy	0.800	0.890	1.2

Policy improves EM by +0.033 and F1 by +0.038 over train-best *while using fewer retrieval calls* (1.2 vs. 2.0). NQ results are flat (EM 0.133 for all methods)—this remains an **api_pilot**, not a final answer-quality claim.

Conclusion

A lightweight contextual bandit can replace fixed-action RAG retrieval, improving retrieval reward on both BEIR datasets while keeping cost explicit. The policy is interpretable (feature-based), transfers across scientific domains, and is strongest on high-difficulty queries where fixed strategies most disagree. DR-based OPE provides stable offline evaluation across logging policies—confirming policy quality without online deployment.

Limitations: retrieval-stage evidence only; policy trained on replayable offline reward table (not real deployed logs); OPE uses simulated behavior logging; semantic features analysis-only; QA pilot small-scale ($n = 30$).